

NATHAN LANDIG

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EDUCATION

University of Michigan

Bachelor of Science in Engineering, Mechanical Engineering; Minor in German

GPA: 3.98 / 4.00

Coursework: Mechanical Behavior of Materials, Fluid Mechanics, Strength of Materials, Design and Manufacturing, Dynamics and Vibrations, Thermodynamics

Ann Arbor, MI

December 2027

Technische Universität Berlin

Study Abroad Research Program

Berlin, Germany

May 2025 – June 2025

- Optimized 3D printing parameters to improve impact resistance of printed models by 500%
- Compiled findings into a research paper and presented results at TU Berlin

ENGINEERING PROJECT EXPERIENCE

Michigan Strength-Augmenting Robotic Exoskeletons (M-STARX)

Mechanical Team Co-Lead

Ann Arbor, MI

September 2024 – Present

- Co-led a 10-engineer sub-team through full-lifecycle design and assembly of a ground-up robotic exoskeleton, managing top-level CAD architecture
- Engineered an adjustable hip-joint mechanism providing 15 in. of total circumferential adjustment, expanding the exoskeleton from single-user to multi-user compatibility across varying pilot heights
- Designed and fabricated custom structural components using 3D printing, waterjet cutting, milling, and turning, applying design-for-manufacturing principles

M-Heal: The Initiative

Electrical Team CAD Specialist

Ann Arbor, MI

September 2025 – February 2026

- Designed and manufactured custom CAD enclosures for electrical hardware and display interfaces on a low-cost neonatal bassinet prototype
- Iterated through 4 design revisions refining fit, tolerances, and usability of a custom interface protecting sensitive microcontrollers and sensors

LEADERSHIP EXPERIENCE

BLUElab EASE

Co-President

Ann Arbor, MI

September 2024 – Present

- Delivered 2 functional prototypes that enabled a pediatric client to navigate school lunch lines independently
- Spearheaded the design of a walker-mounted lunch tray accessory, integrating spatial and safety feedback from Ann Arbor Public Schools physical therapists
- Directed club operations while co-leading a project team focused on developing custom adaptive technology for elementary students with physical disabilities

WORK EXPERIENCE

The Future Innovators Academy

Full-Time STEM Camp Instructor

Ann Arbor, MI

May 2026 – August 2026

- Translated fundamental engineering, physics, and programming concepts into hands-on learning modules for groups of 20+ students
- Enforced strict workshop safety protocols, managed equipment/material allocation, and led risk management

SKILLS

CAD & Analysis: SOLIDWORKS (CSWA), Onshape, Fusion 360, ANSYS Discovery

Manufacturing & Hardware: 3D Printing, DFM, Drafting, Milling, Turning, Waterjet, Circuitry, Microcontrollers

Programming: Python, MATLAB, C++, C#, Java

Languages: English (fluent), German (intermediate)

Affiliations and Interests: Michigan Theme Park Engineering and Design (TPED), Musician (Seventh Street Rock Band)